

popquiz momentum ↕

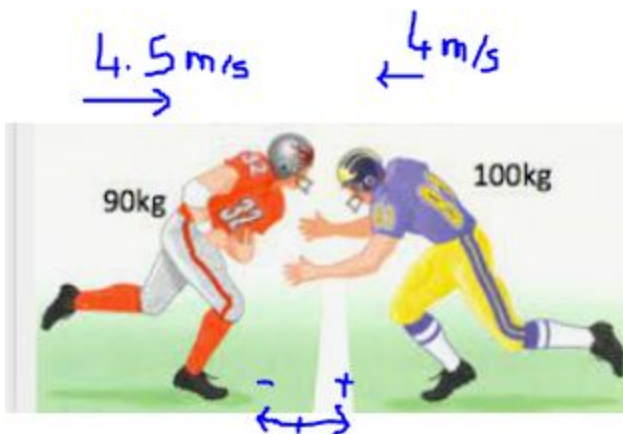
⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 8:02pm

Quiz Instructions



Question 1 14.5 pts



They collide and stick. They will move @ right or left or stop?

refer to previous lecture and video we watched. Which one has the largest momentum

☐

I missed the class

☐

They stop

☐

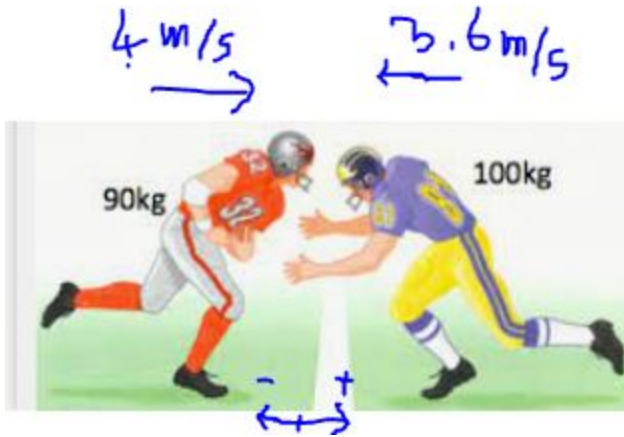
@ left

☐

@ right

⋮

Question 2 14.5 pts



They collide and stick.

☐

They moves at left

☐

They move at right

☐

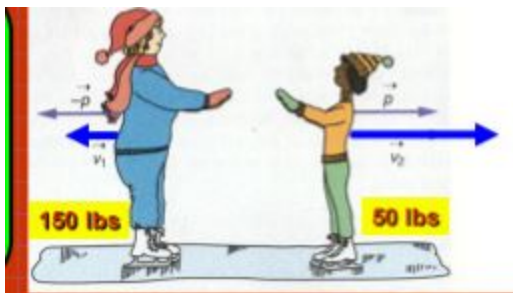
Whaaat?

☐

They stop

⋮

Question 3 14.5 pts



Amy (150 lbs) and Gwen (50 lbs) are standing on slippery ice and push off each other. If Amy slides at 6 ft/s, what speed does Gwen have?

hint: no conversion necessary.

☐

18ft/s

☐

12ft/s

☐

9ft/s

☐

6ft/s



Question 4 14.5 pts

A small car and a large truck collide head-on and stick together. Which one has the larger momentum change?

☐

Not enough information. Missing final speed.

☐

They both have the same momentum change

☐

the truck

☐

the car



Question 5 14.5 pts

A small beanbag and a bouncy rubber ball are dropped from the same height above the floor. They both have the same mass. Which one will impart the greater impulse to the floor when it hits?

Hint: impulse is the change in momentum . see slides.

As discussed in class, this is a bouncing vs crashing situation

- ☐ whaaaat ?
- ☐ both the same
- ☐ the beanbag
- ☐ the rubber ball



Question 6 14.5 pts



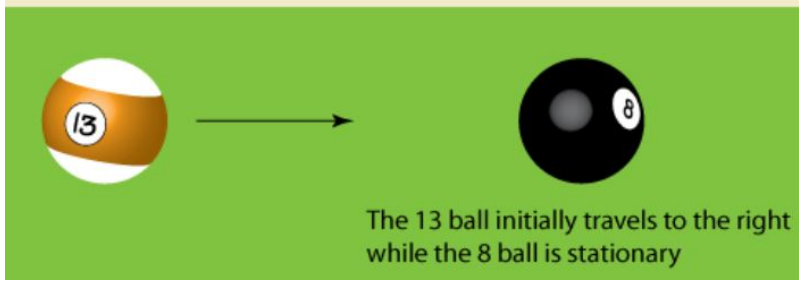
We discussed this demo in class. The basket ball is very very massive relative to the ping pong ball

What is true

- ☐ both the balls come to a stop at collision time.
- ☐ basket ball slows down a tiny bit and ping pong ball bounces off with a much larger speed (than before)
- ☐ basket ball slows down a tiny bit and ping pong ball bounces off with a smaller speed (than before)
- ☐ The 2 balls bounce off each other and move in opposite direction (keeping the same speed as before)



Question 7 14.5 pts



What happens next ? The collision is elastic. They have the same mass. They collide along a line. (1D)

☐

both bounce off each other

☐

I missed the class

☐

13th stops and 8th moves @ right

☐

13th bounces off and 8th stops



Question 8 5 pts

A 400 lbs car is on the surface of the EArth (distance R from center). At a distance $2R$ the weight of the car is

☐

0 lbs

☐

100lbs

☐

800lbs

☐

400 lbs

Not saved

Submit Quiz